

Supporting Evidence – Model Selection Rationale

Prepared for Decision Log 2 | StackEval Benchmark – Professional Research Project

Purpose: This document records the benchmark observations and feasibility checks that shaped the final model set. It exists so the reasoning in Decision Log 2 does not stay abstract. Instead of only saying that the latest API-accessible models were chosen, this document shows the concrete evidence that pushed that decision.

Final model set at a glance

Selected model	Category	Main evidence used	Why it won its slot
GPT-5.4	Frontier proprietary	Version-gap comparison and coding-focused benchmark evidence	The reviewed GPT evidence showed that newer generations can dramatically outperform older ones on reasoning and coding tasks. In addition, across the benchmark panels I analyzed, GPT models consistently appeared in strong positions, especially in coding-oriented evaluations such as SWE Bench. This consistent presence, combined with the observed performance gap between older and newer versions, supported selecting the latest GPT family as the main proprietary model for the benchmark.
Claude Sonnet 4.6	Frontier competitor	Claude comparison and reasoning/coding benchmark evidence	Claude remained highly competitive across the reviewed reasoning and coding panels, making it a strong non-OpenAI comparison point.
Llama 4	Open-source	Speed leaderboard and ecosystem contrast	Llama was selected to represent the open-source category and add a different ecosystem instead of comparing only familiar proprietary systems.
GPT-5.4-mini	Budget / lightweight	StackEval-style coding leaderboard with small-model evidence	The reviewed leaderboard suggested that a smaller OpenAI model can remain highly competitive, which supported adding a lightweight model without expanding the benchmark too far.

1. Why latest versions were preferred over older ones

The first major shift in the selection strategy came from comparing older and newer GPT generations. That comparison made it clear that choosing only familiar or widely used models was not enough. The version gap itself had become a meaningful variable.

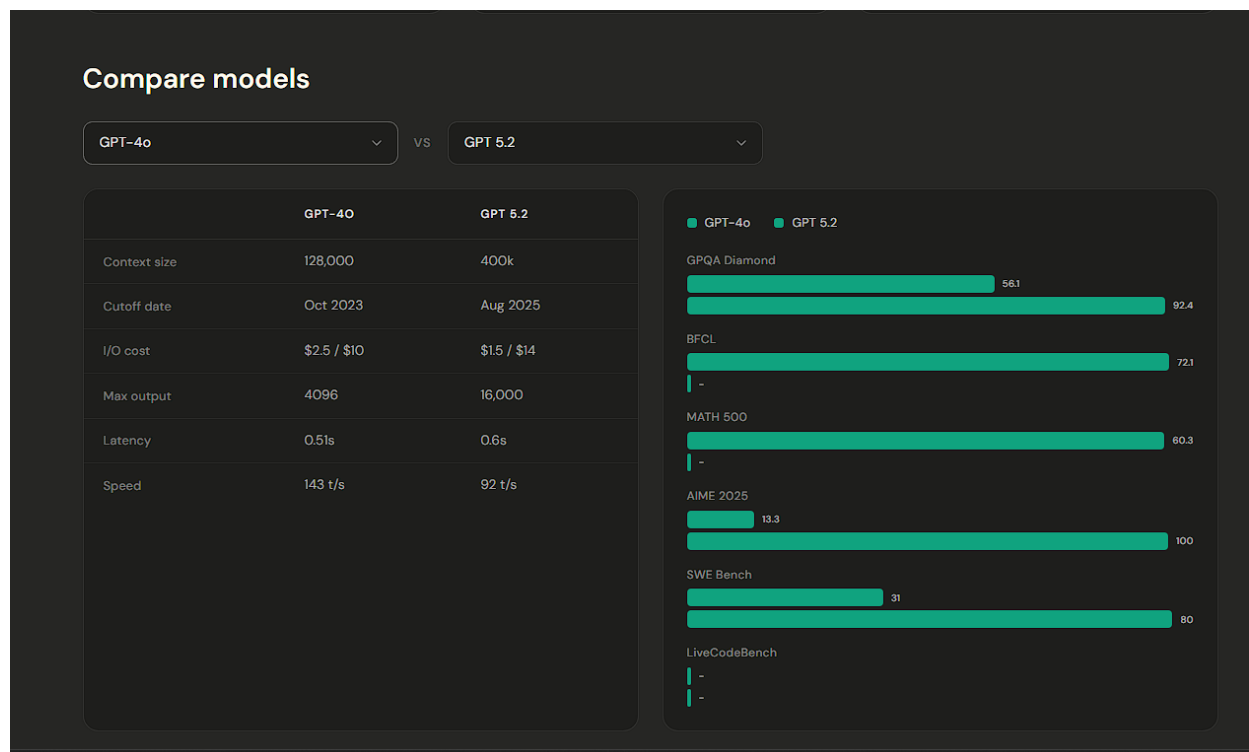


Figure 1. Compared GPT generations reviewed during model selection. The evidence showed a major gap between GPT-4o and GPT 5.2 on AIME 2025 and SWE Bench.

In the reviewed comparison, **GPT-4o scored 13.3 on AIME 2025 and 31 on SWE Bench**, while **GPT 5.2 scored 100 on AIME 2025 and 80 on SWE Bench**. This was not a small incremental change. It showed a large capability jump in both reasoning and coding-related performance.

After seeing this, the selection question changed from “**Which familiar models should I use?**” to “**Which current models best represent each category right now?**”. That is why the strategy moved away from popularity and toward current evidence.

2. Why the GPT family kept the main proprietary slot

For a coding-focused benchmark such as StackEval, coding-specific evidence was more important than general popularity. Across the benchmark panels I reviewed, GPT models consistently appeared in strong positions, especially in coding-related evaluations.

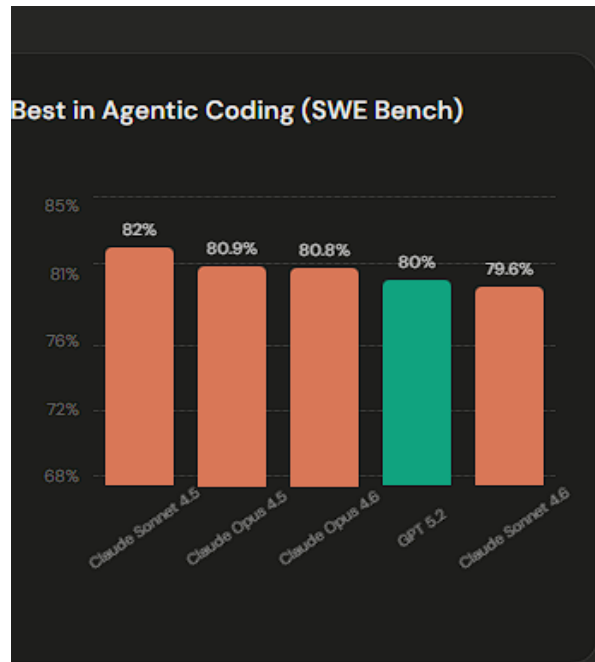


Figure 2. Best in Agentic Coding (SWE Bench) panel reviewed during selection. GPT appeared in the top coding chart, while Gemini did not appear in this consulted view.

In the SWE Bench panel, GPT 5.2 achieved **80% performance** in a coding-oriented benchmark. Since StackEval evaluates models on real developer questions, this type of evidence was directly relevant. In contrast, Gemini did not appear in the top results within the benchmark views I analyzed. This made GPT a more reliable choice for representing the proprietary frontier model in a coding-focused evaluation.

StackEval

Evaluates an LLM's capability to function as a coding assistant by answering a variety of coding-related questions across different programming languages and question types.

#	Name	Provider	Acceptance
1	GPT-4.1	OpenAI	0.986
2	GPT-4.1 Mini	OpenAI	0.984
3	Claude-v4 Sonnet	Anthropic	0.978

Full Leaderboard →

Figure 3. StackEval-style coding assistant leaderboard reviewed during selection. GPT-4.1 and GPT-4.1 Mini appeared with very similar acceptance values.

The decision was further supported by the StackEval-style coding leaderboard. In this view, **GPT-4.1 achieved 0.986 acceptance**, while **GPT-4.1 Mini achieved 0.984 acceptance**. The minimal difference between these values indicated that smaller variants of GPT can still perform competitively in coding tasks. This justified including a lightweight GPT model as part of the benchmark.

In addition to benchmark evidence, GPT models appeared consistently across multiple evaluation panels I reviewed, often standing out in both reasoning and coding-related metrics. This consistency increased confidence that GPT would provide stable and meaningful results in the benchmark.

Finally, I had more prior experience working with GPT models compared to alternatives such as Gemini. While familiarity alone is not a sufficient criterion, it made the experimental process more controlled and interpretable, especially when analyzing model behavior and debugging outputs.

3. Why Claude won the competitor slot

Claude remained in the final set because it repeatedly appeared as a strong competitor in the reviewed reasoning and coding evidence. It represented a different provider while still staying competitive on the kinds of capabilities that matter for StackEval.

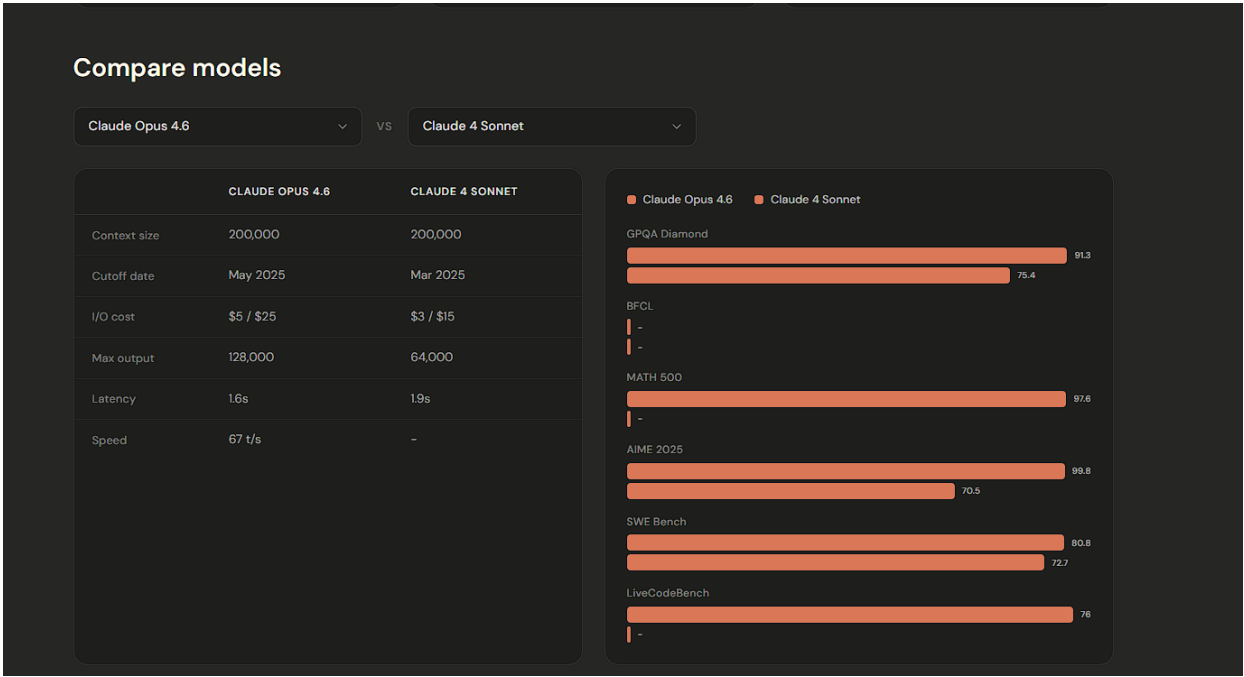


Figure 4. Compared Claude models reviewed during selection. Claude Opus 4.6 showed stronger scores than Claude 4 Sonnet across multiple benchmark areas.

In the compared Claude view, **Claude Opus 4.6 scored 91.3 on GPQA Diamond, 99.8 on AIME 2025, and 80.8 on SWE Bench**. By contrast, **Claude 4 Sonnet scored 75.4 on GPQA Diamond, 70.5 on AIME 2025, and 72.7 on SWE Bench** in the same reviewed comparison.

This was useful because it showed that Claude was not included only because it is a famous model family. The reviewed evidence suggested that Claude remained genuinely competitive in reasoning-heavy and coding-adjacent tasks.

4. Why Llama won the open-source slot

I did not want the benchmark to compare only familiar proprietary systems. A strong model-selection document should also show why one model from a different ecosystem was worth testing.

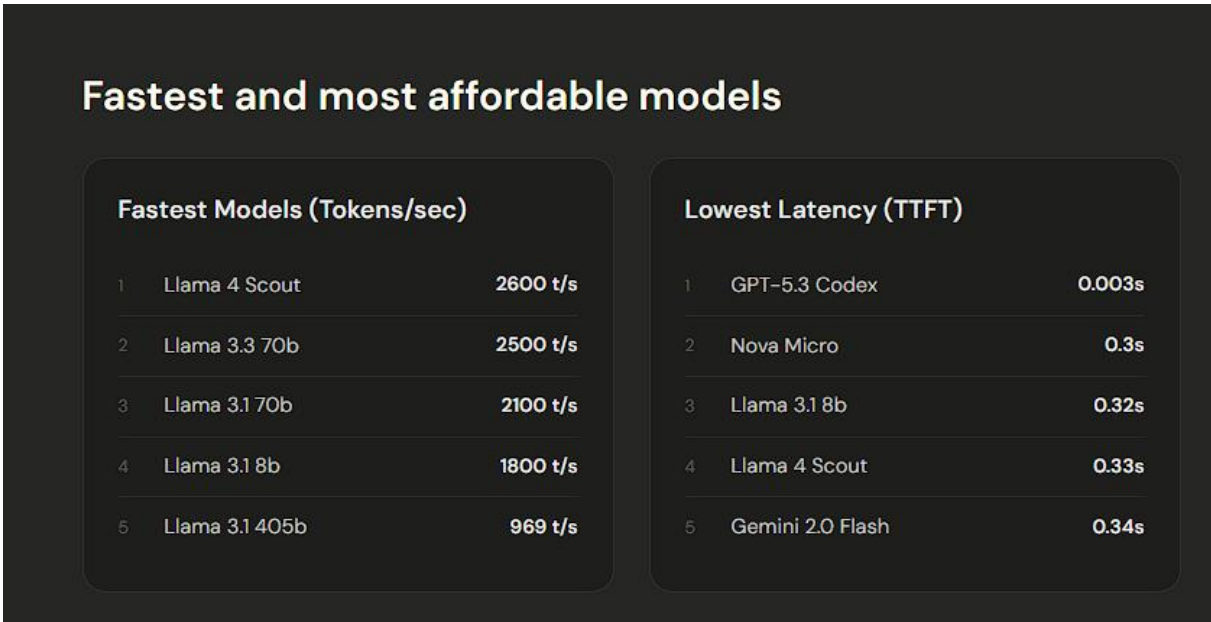


Figure 5. Fastest and most affordable models panel reviewed during selection. Llama 4 Scout appeared as the fastest listed model.

The reviewed speed chart showed **Llama 4 Scout at 2600 tokens/sec**, which made the Llama ecosystem meaningful to include. This supported the open-source slot for two reasons: it covered a different ecosystem, and it added a strong speed contrast to the benchmark.

This part of the choice was also honest on a personal level. I was less familiar with the Llama ecosystem, but the benchmark evidence gave me a real reason to explore it rather than avoid it.

5. Why API feasibility still mattered

Performance alone was not enough. Because Decision Log 3 chose **API-based querying**, the selected models also had to remain usable in an automated benchmark script.

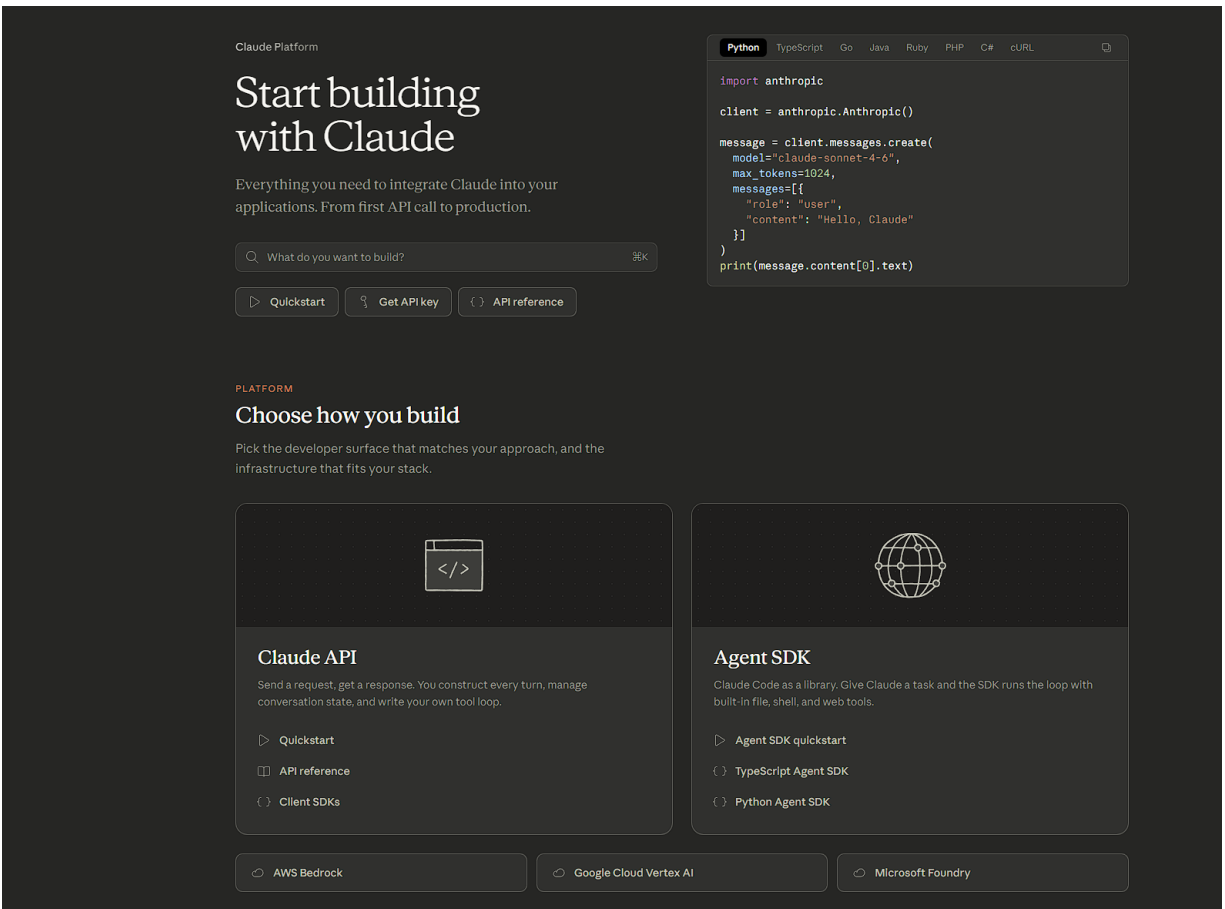


Figure 6. Example API feasibility evidence reviewed during model selection: Claude platform documentation.

So the model-selection process included feasibility checks such as reviewing provider model pages, checking API documentation, and confirming that the models could be reached directly or through a stable API platform. A model that looks strong on a leaderboard but cannot be used reliably in the benchmark would weaken the project design.

6. Additional benchmark view used during selection

A broader leaderboard view was also useful because it showed which model families kept appearing across different benchmark categories, instead of only one task type.



Figure 7. Multi-benchmark leaderboard view reviewed during selection. Claude, GPT, Gemini, and Llama-family models appeared across different capability areas.

This broader view helped confirm that the selected set should not be based on one screenshot alone. Instead, the final benchmark set was shaped by repeated patterns across reasoning, coding, mathematics, and visual reasoning panels.

7. Final conclusion

The final set was chosen by combining **category coverage**, **concrete benchmark evidence**, and **API feasibility**. That changed the selection from a mostly descriptive list into a more defensible methodological choice.

- GPT-5.4 – frontier proprietary model
- Claude Sonnet 4.6 – frontier competitor
- Llama 4 – open-source model
- GPT-5.4-mini – budget / lightweight model

References used in this supporting document

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